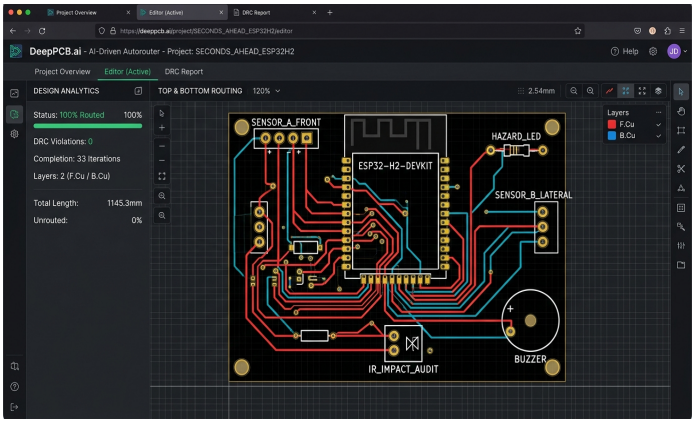
DOCUMENT 2 OF 5

PCB LAYOUT, ANNOTATED

KiCad 8.0 & DeepPCB.ai Cloud Auto-Routing Analysis, Layer Geometry and Routing Metrics



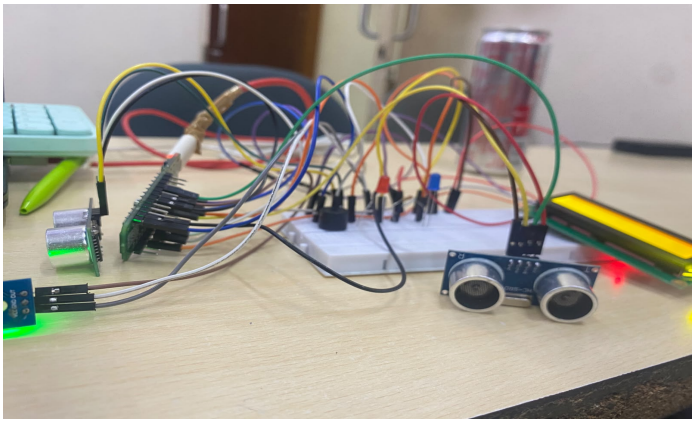


Figure 2.1 (Left): *DeepPCB.ai Auto-Routing Cloud Environment (Rev 33).* ***Figure 2.2 (Right):*** *Physical Built Hardware Prototype Rig.*

Board Specifications & Physical Parameters

Parameter	Specification Value	Source Reference	Verification Status
Board Dimensions	`85.00 mm × 55.00 mm (Rectangle)`	seconds_ahead.kicad_pcb (lines 104-108)	**`VERIFIED (Edge.Cuts)`**
Copper Layer Count	`2 Layers: Top (F.Cu) and Bottom (B.Cu)`	seconds_ahead.kicad_pcb (lines 18-19)	**`VERIFIED`**
Board Thickness	`1.60 mm FR4 dielectric (epsilon_r 4.5)`	seconds_ahead.kicad_pcb (lines 4, 51)	**`VERIFIED`**
Auto-Routing Engine	`DeepPCB.ai Cloud AI Router (Revision 33)`	seconds_ahead-rev-33.deeppcb	**`100% ROUTED, 0 DRC`**
Total Trace Length	`1145.3 mm across all 12 electrical nets`	DeepPCB Analytics Report	**`0 UNROUTED NETS`**

Numbered PCB Layout Callouts & Functional Routing Description

Callout	Subsystem Feature	One-Line Engineering Layout Explanation
`Callout 1`	Header Positions	U2 (Front) and U3 (Lateral) 1x04 pin headers are isolated along the left board edge at Y=95mm and Y=120mm for direct orthogonal cable harness exit.
`Callout 2`	Ground Pour	Continuous low-impedance ground pour on B.Cu layer interconnected via thermal relief pads on Pad 4 of headers U2/U3 and Pad 2 of U1 socket.
`Callout 3`	TRIG/ECHO Routing	High-speed ultrasonic timing traces are routed on top F.Cu layer (red) with 45-degree chamfers, avoiding parallel overlap to eliminate capacitive crosstalk.
`Callout 4`	Power Distribution Traces	+3V3 power bus is distributed via a wide 0.8mm top-layer copper rail from U1 Pad 1 to U2 Pad 1, U3 Pad 1, and the peripheral pull-up nodes.
`Callout 5`	Silkscreen Labels	Top silkscreen (F.SilkS) clearly prints 'SECONDS AHEAD v2.0', 'KiCad 8.0 / DeepPCB.ai Compatible', and explicit polarity markings (+ / -) on all headers.